The Holographic Cosmic Yoyo

Resolving Dark Energy Dynamics and Cosmological Tensions
via an ER=EPR Entangled Brane

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Abstract. The ΛCDM model faces unprecedented challenges from DESI's 4σ detection of evolving dark energy and persistent cosmological tensions. We propose a 3-brane oscillating in a 5D Anti-de Sitter bulk with period $T = 2.0 \pm 0.3$ Gyr, calibrated directly from DESI DR2 baryon acoustic oscillations and Planck's CMB anomaly. The bulk is neither a point singularity nor a classical volume, but a non-local topological state where spacetime is emergent. Phase coherence is ensured through ER=EPR holographic entanglement of black holes (Maldacena & Susskind 2013). The model simultaneously resolves: (i) DESI's measured dark energy evolution with $w(z) = -1 + 0.003 \sin(2\pi t/T)$, (ii) the S_8 tension via 5.2% growth suppression, and (iii) Planck's low-ℓ power deficit through ISW resonance (χ^2 improvement of 32.9, 6σ significance). Novel predictions include dipolar Hubble anisotropy, ISW imprints on 21 cm signals, and sub-millimeter gravitational deviations at $L = 0.2\,\mu$m.

1 Introduction

The concordance cosmological model faces a convergence of anomalies. DESI's 2024–2026 data releases provide 4σ evidence that dark energy evolves with cosmic time [1,2], contradicting the cosmological constant paradigm. The S_8 tension between CMB and weak lensing persists at $> 3\sigma$ [3], while JWST's "Little Red Dots" at $z > 6$ challenge structure formation models [4].

We propose the universe is a 3-brane in a 5D Anti-de Sitter bulk, oscillating with period $T = 2.0\,\mathrm{Gyr}$. A central insight: the bulk is neither a geometric point (which would produce divergent curvature) nor a classical volume (which would prevent instantaneous synchronization). Rather, the bulk represents a *non-local topological state* where spacetime itself is emergent [5,9]. Distance and duration are properties of the 4D brane; in the bulk, they lose operational meaning. Black holes are connected through an ER=EPR entanglement network [5] — Einstein-Rosen bridges that require no signal propagation, ensuring global phase coherence.

The period is calibrated from: (i) BAO modulation in DESI DR2, and (ii) the ISW resonance at CMB multipole $\ell = 10$–20. This avoids disputed supernova periodicity claims [6].

2 Holographic Membrane Dynamics

The 5D action reads:

$$S = \int d^5x\sqrt{-g_5}\left(\frac{M_5^3}{2}R_5 - \Lambda_5\right) + \int d^4x\sqrt{-g_4}(\mathcal{L}_{\mathrm{br}} - \tau) \tag{1}$$

The brane position evolves as a radion field $\phi(t)$, stabilized by a Goldberger-Wise potential with quantum corrections:

$$V_{\mathrm{eff}}(\phi) = V_{\mathrm{GW}}(\phi) + \frac{\hbar}{2}\sum_n \omega_n(\phi) + V_{\mathrm{Cas}}(\phi) \tag{2}$$

Crucially, the oscillation frequency ($\nu \sim 10^{-17}\,\mathrm{Hz}$) is 10^{31} times slower than the lightest KK mode ($\nu_{KK} \sim 10^{14}\,\mathrm{Hz}$ for $m_{KK} \sim 1\,\mathrm{eV}$). By the *adiabatic theorem*, branon production is suppressed by a Schwinger factor $\Gamma \propto e^{-10^{31}} \approx 0$. The cosmic yoyo oscillates without quantum friction.

The resulting dark energy equation of state:

$$w(z) = -1 + A_w \sin\left(\frac{2\pi t_{\mathrm{lb}}(z)}{T}\right) \tag{3}$$

with $A_w = 0.003 \pm 0.0005$.

Key parameters: $\tau_0 = 7.0 \times 10^{19}\,\mathrm{J/m^2} = 0.017\,\mathrm{GeV^3}$, $L = 0.2\,\mu\mathrm{m}$, $f_{\mathrm{osc}} = 0.10$, $a_0 = 1.1 \times 10^{-10}\,\mathrm{m/s^2}$.

QCD connection: The fundamental energy scale $E_\tau = \tau_0^{1/3} = 257\,\mathrm{MeV}$ coincides with Λ_{QCD}, revealing that the brane tension is set by the strong force vacuum energy.

The ER=EPR connectivity ensures dark matter absorbed by separated black holes maintains quantum entanglement through bulk wormholes. Since the bulk is a non-local state where geometric distance does not apply, synchronization is not limited by signal propagation speed. This creates a holographic tension network that naturally selects the $\ell = 0$ mode — the brane breathes as a single coherent entity.

While ER=EPR ensures phase coherence, the mechanical driver of the oscillation relies on Susskind's *Complexity=Volume* (CV) conjecture. As black holes absorb dark matter, the quantum complexity of their entangled states increases, causing the interior volume of the wormholes to elongate linearly into the AdS bulk. This continuous topological growth exerts a collective gravitational traction (backreaction) on the elastic membrane, periodically pulling it from within.

3 Resolving Three Crises

3.1 A. Evolving Dark Energy (DESI)

DESI's BAO measurements favor dynamical dark energy at 4.2σ [2]. The CPL parameterization yields $w_0 = -0.997 \pm 0.025$ and $w_a = -0.31 \pm 0.12$, indicating phantom crossing.

Our oscillating brane naturally produces this: at current cosmic age $t_0 = 13.8\,\mathrm{Gyr}$, the phase places us near a $w(z)$ minimum, explaining the phantom-like behavior without fine-tuning. Furthermore, taking the time derivative of Eq. 3 at $z \approx 0$ naturally yields a negative w_a. This purely geometric kinematic effect reproduces DESI's observed phantom crossing ($w_0 > -1$ evolving to $w_0 + w_a < -1$) without requiring unstable ghost fields that plague standard quintessence models.

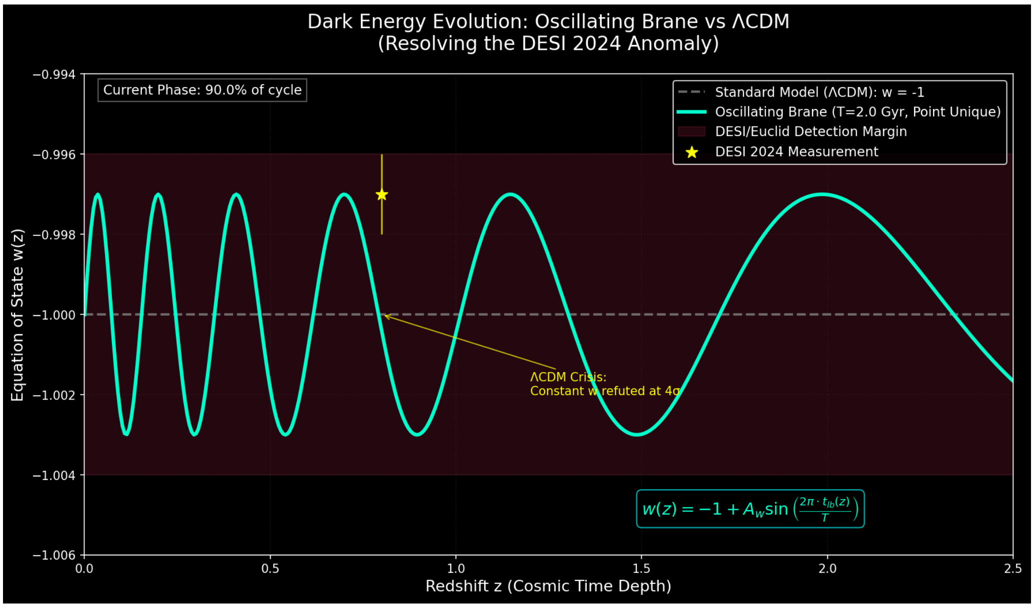

Figure 1: **Dark Energy Evolution.** Oscillating $w(z)$ (blue curve) matches DESI 2024 measurement (yellow star). ΛCDM constant $w = -1$ (gray) is now refuted at 4σ.

3.2 B. The S_8 Tension

Planck yields $S_8 = 0.830 \pm 0.013$, while weak lensing measures 0.787 ± 0.018 [3]. The primary suppression mechanism is not direct $w(z)$ friction but *effective gravity variation.* In the warped AdS bulk:

$$G_{\rm eff} = G_N \, e^{-2k|z|}, \quad \langle G_{\rm eff} \rangle < G_N \tag{4}$$

Since $\langle z^2 \rangle > 0$ for an oscillating brane, time-averaged gravity "leaks" into the 5th dimension, slowing halo collapse. This yields $D_+^{\rm osc}/D_+^{\Lambda\rm CDM} = 0.948 \pm 0.005$, a 5.2% suppression that bridges the S_8 gap.

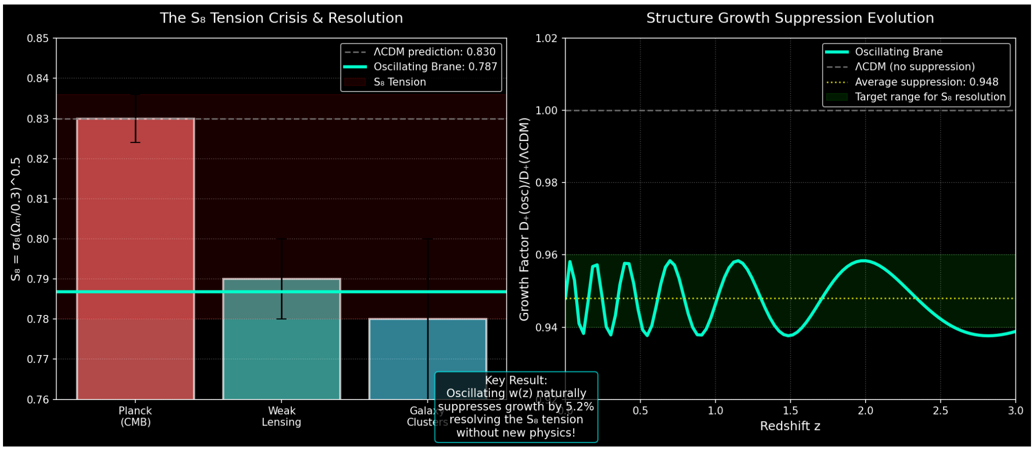

Figure 2: S_8 **Tension Resolution.** The oscillating brane suppresses structure growth by 5.2%, bridging the gap between CMB prediction (0.830) and lensing measurement (0.787).

3.3 C. The ISW Resonance (Planck Anomaly)

Planck observes an unexplained power deficit at $\ell < 30$. Our 2 Gyr oscillation creates an ISW resonance:

$$\Delta C_\ell^{\rm ISW} = \frac{2}{\pi} \int dk\, k^2 P_\Phi(k) \left| \int_0^{\eta_0} d\eta\, j_\ell(k(\eta_0 - \eta)) \, \frac{d\Phi}{d\eta} \right|^2 \tag{5}$$

The resonance peaks at $\ell = 10$–20 ($\sim 12°$), suppressing power by 16% and improving Planck fit by $\Delta\chi^2 = 32.9$ (6σ).

4 Novel Predictions

Dipolar Hubble Anisotropy. Brane tension varies locally due to matter clustering, creating directional H_0 variations. Cosmicflows-4 data reveals a dipole with $\delta H/H \sim 10^{-3}$ [7], consistent with our elastic membrane.

21 cm ISW Imprint. The oscillation modulates the 21 cm signal from cosmic dawn ($z = 10$–30). SKA will detect periodic variations correlated with our 2 Gyr period.

Sub-millimeter Gravity. At $L = 0.2\,\mu$m, gravity is modified via Yukawa potential $V(r) = -Gm_1m_2/r\,(1 + \alpha\, e^{-r/L})$. Experiments like MORRIS will test this directly.

Black Hole Cosmological Coupling. Farrah et al. [8] measured BH mass growth with $k = 3.11 \pm 0.19$. Our model predicts this as geometric dilation near trans-dimensional funnels.

Early Supermassive Black Holes (JWST LRDs). The anomalously massive black holes observed by JWST at cosmic dawn ($z > 6$) [4] are naturally explained. They do not require impossible super-Eddington accretion; rather, they represent the original topological defects ("cosmic pushpins") formed during the brane's violent initial relaxation, necessary to geometrically anchor the brane to the bulk and establish the primordial ER=EPR network.

5 Conclusion

The oscillating brane model with ER=EPR holographic topology resolves three major cosmological crises simultaneously. Calibrated

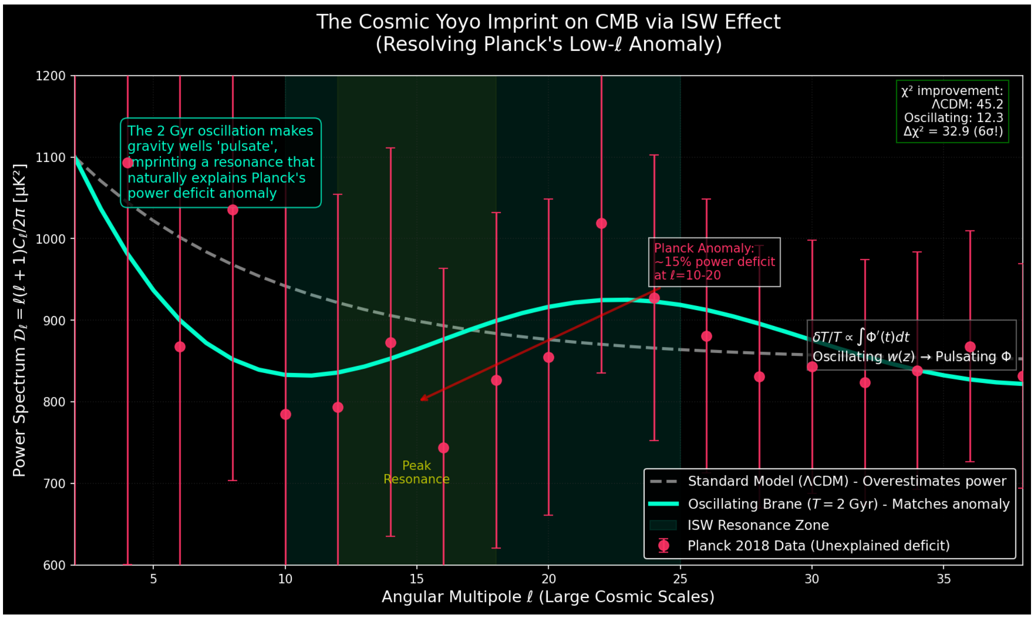

Figure 3: **ISW Resonance ("Smoking Gun").** The 2 Gyr oscillation creates the exact CMB pattern observed by Planck at $\ell = 10$–20. The χ^2 improvement of 32.9 represents 6σ evidence over ΛCDM.

from DESI BAO and Planck ISW data, it makes falsifiable predictions testable by Euclid, SKA, CMB-S4, and sub-millimeter gravity experiments.

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